Check your understanding

1. Describe the relationship between the length of a wire and its resistance. [1 mark]

   .....

2. Use your graph to state whether this relationship is **proportional** or **directly proportional**.
   Justify your answer. [2 marks]

   .....

   .....

3. By adding a bulb to your circuit, you increase the resistance.
   Explain why this stops the wire from overheating. [2 marks]

   .....

   .....

Exam-style questions

1. State the units for potential difference, current and resistance. [2 marks]

   a. Potential difference (V) .....

   b. Current (I) .....

   c. Resistance (R) .....

2. State the equation that links potential difference, current and resistance. [1 mark]

   .....

3. Current is affected by the resistance of a wire.
   Describe what happens to the current as the resistance of a wire is increased. [1 mark]

   .....

4.2.1.3 Resistance

Check your understanding

1. As the length of the wire increases, the resistance also increases. [1]
2. The relationship is directly proportional [1]
   as it is a straight line that passes through the origin. [1]
3. If the resistance is higher, then the current will be lower. [1]
   A lower current passing through a wire will mean it doesn't get as hot / less energy will be transferred as thermal energy. [1]

Exam-style questions

1. **a.** Potential difference (V) = V (volts)
   b. Current (I) = A (amperes, amps)
   c. Resistance (R) = Ω (ohms) [2]
   2 marks for all three correct;
   1 mark for two or one correct
2. $V = IR$
   resistance = potential difference ÷ current
   1 mark for either version [1]
3. As resistance increases, current decreases. [1]